

1 Boundless Skies Node Guidelines

Version: 1.0

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1.1 1. Mission Alignment

- Primary goal: Maximum accessibility for people with disabilities — zero/low barrier to entry.
- “Bring Your Own” hardware is always free (software-only).
- Pre-configured nodes are a convenience/revenue option (10-15% markup to cover servers, ops, and sponsor free nodes).
- All nodes use official ZWO Seestar ALPACA (safe, cross-platform Python).

1.2 2. Supported Hardware Policy

- **Sold / Preloaded Nodes:** Seestar family only (S30 Pro, S50, etc.) for reliability.
- **Bring Your Own:** Allowed for free software path. Primarily Seestars; ALPACA-compatible scopes can apply for manual approval.
- Station Mode required: Scope joins home WiFi (same network as Node computer) for ALPACA control + SMB image sharing + internet.

1.3 3. Node Builder Flow (Rolls-Royce Style Configurator)

Sleek, dark space-themed UI/UX: - Mobile-first wizard with live preview, big visuals, accessibility (VoiceOver, large taps, audio descriptions). - Starts with quick Hardware Inventory Quiz → personalized recommendation. - Persistent live summary panel (price, science impact, uptime estimate). - Hero “Complete Autonomous Node” preset. - Ability to deselect items on review screen.

Steps: 1. **Hardware Quiz** (detect what user already has). 2. **Telescope:** - Recommended: Seestar S30 Pro / S50. - Tiered options with science impact scores. - “I already own one”. 3. **Computer:** - Raspberry Pi 5 (primary) or Pi 4B (budget). - Mac Mini (premium). - “I already have one”. 4. **Addons / Bundles** (modular, collapsible): - **Power & Autonomy:** Smart Power Box, SwitchBot, UPS/Battery, Solar Power Kit. - **Protection:** Minidome + weather sensors, Dew Heater (optional). - **Connectivity:** WiFi Extender / Mesh. 5. **Review & Checkout:** Deselect items, see impact, markup explanation.

1.4 4. Complete Autonomous Node Preset

Flagship option: - Seestar (S50 recommended) - Raspberry Pi 5 preloaded - Full Autonomy Pack (Power Box + SwitchBot + UPS + Solar) - Protection Pack (Minidome + sensors) - WiFi Extender - Users can customize by removing items (e.g., skip dew heater or dome).

1.5 5. Rebranding (Sleek & Professional)

- **Hardware:** Matte black/gunmetal vinyl wraps, custom stickers/QR plates with Boundless Skies logo + subtle starfield/accessibility motif. Cost: \$20–60 per unit.
- **Software:** Custom boot splash, wallpaper, Node Agent branding.
- **Packaging:** Premium constellation-themed box, activation card, quick-start guide (large print + audio).
- Feels cohesive and premium (NASA + Apple aesthetic).

1.6 6. Activation & Standardization

- Every node (free or purchased) gets a unique **Node Activation Code** (e.g., BS-ELI-2026-XYZ).
- Mailed physical card or digital PDF with Node ID, equipment checklist, QR for setup.
- Auto-registers on first boot of Node Agent.

1.7 7. Preloading Process

- All sold hardware pre-flashed with Node Agent (Phase 0+), tested, Boundless Skies branded.
- Includes config wizard, local web UI, safety rules.

1.8 8. Pricing Examples (Approximate, 2026)

- Software-Only: \$0 (or donation)
- Basic Node (Seestar + Pi 5): \$850–\$1,050
- Protected Node: \$1,100–\$1,400
- Complete Autonomous Node: \$1,600–\$2,200 (customizable)

Markup funds ops + free/sponsored nodes for disabled members.

1.9 9. Technical Integration

- Node Agent controls addons via MQTT/Home Assistant bridge (power, sensors, etc.).
- Safety: Heartbeats, auto-park on issues, weather integration.
- Ties into full tech stack (ALPACA, photometry, cloud scheduling).

1.10 10. Future Expansions

- Tier 2 filtered nodes.
- NINA/EKOS/ASIAIR support (Phase 4).
- More addons based on demand.

This document captures all discussions from the Node Options conversation. It keeps the experience simple, accessible, and premium.

Next Actions: Finalize Node Agent skeleton, build the configurator UI, source SKUs.